LRU Page Replacement Algorithm

Write a C program to implement the LRU Page Replacement Algorithm. The program should accept a reference string and the number of frames. For each page request, identify whether it results in a Hit or a Fault. If a Fault occurs and the frames are full, replace the page that was accessed least recently. Display the frame status at each step and calculate the total number of Page Faults and Hits

Input Format

1. The first line of input contains an integer representing the number of pages in the reference string. \
2. The second line contains the page reference string as a sequence of integers.
3. The third line contains an integer representing the number of available frames.

Output Format

The output displays the total number of Page Faults and the total number of Page Hits after processing the entire reference string using the LRU page replacement algorithm.

Sample Input 1

5

1 2 3 4 5

5

Sample Output 1

Enter number of pages in reference string:

Enter the reference string:

Enter number of frames:

Total Page Faults: 5

Total Page Hits: 0

CODE:

```
#include <stdio.h>

int main() {
   int n, frames;

   printf("Enter number of pages in reference string: \n");
   scanf("%d", &n);

   int  pages[n];     printf("Enter   the   reference
   string: \n"); for(int i = 0; i < n; i++) {
      scanf("%d", &pages[i]);
   }

   printf("Enter number of frames: \n"); scanf("%d",
   &frames);

   int frame[frames], time[frames];

   // Initialize for(int i = 0; i < frames;
   i++) {
      frame[i] = -1; time[i] = 0; } int pageFaults =
   0, pageHits = 0, counter = 0;

   for(int i = 0; i < n; i++) {
      int found = 0;

      // Check for Hit for(int j = 0; j
      < frames; j++) {
         if(frame[j] == pages[i]) {
         found  =          1;
         pageHits++; counter++;
            time[j]              =
            counter; break;
         }
      }

      // Page Fault if(!found)
      {  int lruIndex =
         0;
```

```
        for(int j = 1; j < frames; j++) {
          if(time[j] < time[lruIndex]) {  lruIndex
             = j;
          }
        }

        counter++;
        frame[lruIndex]        = pages[i];
        time[lruIndex] = counter;
        pageFaults++;
      }
    }

    printf("\nTotal Page Faults: %d\n", pageFaults); printf("Total
    Page Hits: %d\n", pageHits);

    return 0;
}
```

OUTPUT: